

Graduation Showcase: The Junior Innovator's Guide

Mastering Design Thinking, Functional Innovation & Technical Creativity

Welcome to the World of Design!

Did you know that you are already a designer? Every time you find a way to make something work better or solve a daily problem, you are using **Design Thinking**.

What is a Gadget? A gadget is a functional tool created to solve a specific human problem—whether simple (like a manual pump) or advanced (like code and software!).

Today's Mission

Learn how real-world heroes use simple materials and code to solve big problems, and prepare your own project for the Graduation Showcase using the Design Loop!

1. Functional Design & Real-Life Innovators

Design isn't just about how things look—it's about **how they work!**

Mansukhbhai Prajapati (Mitticool)

Created a clay refrigerator that keeps water, milk, and vegetables cool naturally using water evaporation—**zero electricity needed!**

Remya Jose (Pedal Washing Machine)

As a school student, she created a pedal-powered washing machine from bicycle parts. It washes clothes while exercising, saving time and power!

2. Student Showcase Ideas: Showcase What You Know!

At our Graduation Showcase, students can present physical prototypes OR digital projects they have previously learned about to demonstrate their understanding!

Project Category	What You Can Bring & Showcase
Software & Apps (Code)	Showcase a Flask Web App or a customized React Component you built (e.g., an interactive quiz app, task planner, or dashboard).
Finance & Business Plans	Present a project Finance Plan or budget tracker (e.g., product pricing model, mini-business pitch, or savings tracker).
Home & Eco Gadgets	Mini clay-based cooling boxes (inspired by Mitticool), zero-electricity preservers, or simple eco-friendly water dispensers.
Space-Saving & Desk Tools	Multi-tiered organizers (inspired by the stacked Idli Stand), anti-roll pencil holders, or collapsible desk stands.

3. The Design Loop: THINK, MAKE, SHARE

STEP 1: THINK

Find a problem! Ask: *"What is hard to do?"* or *"How can we make this task faster or easier?"*

STEP 2: MAKE

Build a **Prototype** using cardboard, code, or charts. It's a rough working model to test your ideas!

STEP 3: SHARE

Prepare a 30-second pitch explaining: 1. Problem, 2. Solution, 3. What makes your design unique!

Practice Worksheet: Future Innovators Challenge

Complete the 5 practice questions below to prepare for your Graduation Showcase presentation!

Question 1: Defining Functional Design

What is **Functional Design**, and how does a daily object like an *Idli Stand* or a *Cycle Pump* demonstrate it?

Example Answer: Functional Design means creating something specifically to solve a practical problem, not just to look pretty. For example, an *Idli Stand* uses vertical stacking to cook multiple idlis at once while saving limited kitchen space!

Question 2: Learning from Real Innovators

Choose either **Mansukhbhai Prajapati** (Mitticool) or **Remya Jose** (Pedal Washer). Describe the problem they saw and how their functional design solved it without using electricity.

Example Answer: Remya Jose noticed that washing clothes by hand took a long time and was tiring. She built a pedal-powered washer from bike parts so people could wash clothes easily while exercising, saving both electricity and time!

Question 3: Application & Prior Project Showcase (Software / Finance / Hardware)

You can bring a project you have already learned about to the showcase—such as a **Flask web app**, a **React component**, a **finance plan**, or a physical gadget! Which project will you bring or design, and what problem does it solve?

Example Answer: I will showcase my *React Weather Component* that I built! It solves the problem of students forgetting their umbrellas by showing instant weather updates before school.

Question 4: The Design Loop & Prototyping

What is a **Prototype**? Name two simple materials (or software tools) you can use during the **MAKE** step to build your project model.

Example Answer: A prototype is a draft or rough working model of an invention used to test ideas. To build one, I can use cardboard boxes, rubber bands, or code templates!

Question 5: Crafting Your 30-Second Showcase Pitch

When you **SHARE** your project with parents and teachers, what 3 key details should you include in your 30-second presentation pitch?

Example Answer: In my pitch, I should state: 1. What real-world problem I am solving, 2. How my gadget/app works to solve it, and 3. One special feature I am proud of!